

Experimental Design and Analysis of Variance
Project Proposal

Maximizing Concentration

: The Effect of Learning Methods and Ambient Noise
on Memorization Efficiency

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01

Research Question

Which combination of learning output & environment optimizes concentration and memorization accuracy?

01

Research Question

Why it is Interesting?

- **Scientific Approach to Learning:** Finding the Best Study Mode for Any Environment
- **Analyzing Interaction:** Moving Beyond Simple Averages

02 Preliminary Design Details

- **Structure:** 2 x 3 Factorial Design
- **Factor A:** Vocalize vs Write
- **Factor B:** Silence vs White Noise vs K-POP with Lyrics
- **Response Variable (Y):** Accuracy rate
(number of words recalled)
- **Substantiality:** 36 total observations
(6 treatments×2 subjects×3 replicates)

03 Feasibility & Controls

Resources

- Earplugs
- YouTube white noise videos
- French wordbook
- Pen
- A4 sheet
- K-POP with Lyrics
- Timer



The Easiest Self-Study French for Beginners,
Supplement Handbook

03 Feasibility & Controls

Time Management

Memorizing 20 random French words within a 5-minute time limit

- Total Trials: 36 Replications
- Total Time: 180 Minutes

Technical Controls

- **Randomization:** The order of 36 sessions will be completely randomized
- **Blocking:** Using "Person" as a blocking factor to statistically control for individual differences in baseline memory

03 Feasibility & Controls

Statistical Analysis

- **Model:** Two-way ANOVA
- **Effect:** Main Effects & Interaction Effect
- **Method:** SAS/R Programming
- **Goal:** Optimal Combination Discovery

Project Timeline

Week 11

Design Finalization & Experiment Launch

Week 12

Project Progress Report

**Week
13 & 14**

Data Completion & Statistical Analysis

Week 15

Design Finalization & Experiment Launch

Thank you!